

Metamorphism of Siliceous Carbonates: The CMS System & Protolith Diagnosis

Authoritative Reference: John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., Chapter 29: Metamorphism of Calcareous & Ultramafic Rocks)

Comprehensive Chapter Masterclass: Metamorphism of Carbonate Rocks

The CaO–MgO–SiO₂–H₂O–CO₂ (CMS-HC) System, Decarbonation Sequences & T-X(CO₂) Phase Equilibria

1. The Chemical System of Carbonate Metamorphism

Sedimentary carbonate rocks range from pure limestone (CaCO_3) and dolostone ($\text{CaMg}(\text{CO}_3)_2$) to **siliceous dolomitic limestones** and **marls** containing varying amounts of detrital quartz (SiO_2) and clay minerals (illite, smectite).

The phase equilibria of these rocks are modeled within the fundamental five-component system $\text{CaO} - \text{MgO} - \text{SiO}_2 - \text{H}_2\text{O} - \text{CO}_2$ (CMS-HC). When quartz is present, progressive metamorphism drives a series of dramatic **decarbonation and mixed-volatile reactions**, liberating gaseous CO_2 (and H_2O) and forming a sequence of calcium-magnesium silicates.

"Metamorphosed calcareous rocks are derived from carbonate-rich sedimentary precursors... Metamorphosed siliceous dolomites are particularly interesting because the reactions that occur during progressive metamorphism produce a sequence of Ca-Mg silicates that serve as sensitive indicators of metamorphic grade and fluid composition."

— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., Chapter 29, p. 670)

conditions. As a result, they have received more attention than their limited occurrence would otherwise justify. At the incipient stages of metamorphism, impure carbonates consist of a mixture of carbonate minerals (calcite, dolomite, and ankerite are most common), perhaps some chert, and detrital material such as fine quartz, feldspar, and hydrous minerals (usually clays). Metamorphism of these rocks is interesting because it involves several devolatilization reactions (typically both decarbonation and dehydration reactions) dependent not only on P , T , and the rock composition, but also on the composition of the associated pore fluid. To understand the metamorphism of these rocks we will have to adopt an approach that deals with these fluids as $\text{H}_2\text{O}-\text{CO}_2$ mixtures.

In this chapter we will concentrate on impure carbonates in which the major initial constituents are dolomite, calcite, and silica (chert, detrital quartz, or SiO_2 in solution). Such rocks correspond nicely to the simple five-component system: $\text{CaO}-\text{MgO}-\text{SiO}_2-\text{H}_2\text{O}-\text{CO}_2$ (***CMS-HC***). When projected from H_2O and CO_2 , the triangular *CMS* chemographic system depicts all the major minerals. Other components may also be present, typically FeO , Al_2O_3 , K_2O , and Na_2O , but in most carbonate rocks they are present in only minor amounts and do not significantly alter the parageneses.

Figure 1 illustrates the *CMS* system and the minerals that typically develop in metamorphosed siliceous carbonates. A comprehensive analysis of metamorphism of these rocks requires that we consider the reactions between

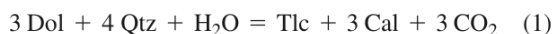
Figure 29-1 from John D. Winter (p. 671): Chemographics in the $\text{CaO}-\text{MgO}-\text{SiO}_2$ (*CMS*) system showing the positions of Calcite (Cal), Dolomite (Dol), Quartz (Qtz), Talc (Tlc), Tremolite (Tr), Diopside (Di), Forsterite (Fo), and Wollastonite (Wo).

2. The Progressive Mineral Sequence in Siliceous Dolomites

As temperature rises from greenschist through amphibolite to granulite facies, the initial sedimentary assemblage **Calcite + Dolomite + Quartz** undergoes sequential reactions as portrayed in Winter's T - X_{CO_2} phase diagram:

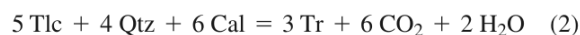
Figure 3 is an isobaric T - X_{CO_2} phase diagram calculated at 0.1 GPa using Rob Berman's TWQ program, based on an internally consistent set of thermodynamic data extracted from experiments at high temperature and pressure. 0.1 GPa is equivalent to a depth of approximately 3 km, suitable for an example of contact metamorphism.

The assemblage **Dol + Cal + Qtz** is stable in siliceous carbonates at temperatures ranging from sediments to the lowest metamorphic grades in Figure 3. As shown in the corresponding compatibility diagram (Figure 4a), this mineral assemblage develops in any bulk composition in the shaded (carbonate) half of the triangle (the right half covers ultramafic rocks). The first metamorphic reaction that occurs (and the temperature at which it takes place) depends on X_{CO_2} of the pore fluid. $X_{\text{CO}_2} < 0.7$ in most situations (the value of invariant point A in Figure 3), so the first isograd is typically:



This reaction is of type 4, consuming H_2O and liberating CO_2 as temperature increases. It involves a tie-line flip in the *CMS* system whereby the low-temperature Dol-Qtz tie-line is replaced by the higher-temperature Tlc-Cal tie-line (compare Figures 4a and b). This reaction introduces **talc** into many metacarbonate rocks. Note in Figure 4b that virtually all dolomitic marbles above the isograd fall into the Dol-Tlc-Cal sub-triangle. Very Ca- and Si-rich carbonates fall in the Cal-Tlc-Qtz sub-triangle. Let's assume that we are dealing with a common siliceous dolostone, in which dolomite is the dominant phase, followed by calcite and finally quartz. How Reaction (1) behaves and what happens next depend on the physical dynamics of the pore fluid, whether open-system or

1.1.1 OPEN-SYSTEM BEHAVIOR If the system is *open* (porous and permeable), the fluid composition may be controlled externally and remain essentially constant, so a traverse up metamorphic grade (toward the igneous contact) follows a vertical path on the T - X_{CO_2} diagram. If we assume that X_{CO_2} of the intergranular fluid phase is initially equal to 0.5, the path is illustrated as the “externally controlled fluid” arrow rising along $X_{\text{CO}_2} = 0.5$ in Figure 3. In such a situation, Reaction (1) is isobarically invariant, because there are $C = 3$ inert components (H_2O and CO_2 are mobile) and $\phi = 4$ coexisting phases, so, according to the phase rule at constant P , $F = 3 - 4 + 1 = 0$. As a result, the temperature remains at that of the isograd, as heat is consumed by the reaction, until one reacting phase is used up. Because we are addressing a dolomitic carbonate with silica impurity, $\text{Dol} \gg \text{Qtz}$, so quartz will be consumed first. Once the quartz is gone, the Dol + Cal + Tlc assemblage is univariant and the temperature can rise again. In this situation the temperature will rise until the system intersects the curve for the reaction:



This reaction involves the appearance of **tremolite** inside the Tlc-Qtz-Cal sub-triangle in the *CMS* system (Figure 4c). This is the “tremolite-in” isograd for many siliceous limestones (shown in Figure 2) and the more SiO_2 -rich dolostones, but most of the shaded area of dolomitic bulk compositions is still in the Dol-Tlc-Cal field, a field in which Reaction (2) does not occur because one reactant (quartz) is missing.

When Reaction (2) runs to completion in appropriate metacarbonates, either talc or quartz will be consumed. Noting where most dolomitic marbles plot in Figure 4,

Figure 29-3 from John D. Winter (p. 672): Isobaric T - X_{CO_2} phase diagram for siliceous dolomites calculated at $P = 0.1 \text{ GPa}$ showing the progression of decarbonation reactions with increasing metamorphic grade.

Reaction 1: Appearance of Diopside (Medium-to-High Grade)

At amphibolite facies conditions, dolomite reacts with quartz (or earlier tremolite + calcite) to generate the calcic pyroxene **diopside**:



Reaction 2: Appearance of Forsterite (High Grade)

In quartz-depleted, dolomite-rich bulk compositions, diopside and dolomite react to form Mg-rich olivine (**forsterite**):



Notice that forsterite in marbles coexists stably with calcite, but is strictly isolated from free quartz by the intervening diopside tie-line!

Reaction 3: Appearance of Wollastonite (Highest Grade / Granulite Facies)

At peak temperatures and low X_{CO_2} (or very high T in granulites), calcite and quartz directly react to produce the pyroxenoid **wollastonite**:



Reaction 4: K-Feldspar from Illitic Clays

In impure dolomites containing minor detrital illite/potassium clay, phlogopite forms at lower grades and breaks down at granulite facies to generate **K-feldspar**:



Therefore, an assemblage containing **calcite + dolomite + quartz + diopside + wollastonite + K-feldspar + forsterite** represents a classic, high-grade, partially equilibrated **siliceous dolomitic limestone (carbonate rock protolith)** undergoing granulite-facies metamorphism!



Official Exam Question Statement & In-Depth Critical Analysis

Directorate General of Mines Safety (DGMS) 2019 · Junior Scientific Officer Exam · Question 8

Question Statement

Q.8: Presence of calcite, dolomite, quartz, diopside, wollastonite, K-feldspar, forsterite in a regionally metamorphosed rock, under granulite facies suggests which one of the following types of protoliths?

(a) Mafic

(b) Pelitic

(c) Carbonate rock

(d) Ultramafic

Correct Answer: (c) Carbonate rock



The combination of carbonate minerals (calcite, dolomite) with calcium-magnesium silicates (diopside, wollastonite, forsterite) uniquely fingerprints an impure siliceous dolomitic limestone protolith.

Exhaustive Option-by-Option Critical Analysis

(c) Carbonate rock

CORRECT

Why it is uniquely correct: The mineral assemblage listed in the question is the textbook signature of a **siliceous dolomitic limestone or marl** (Winter, 2010, Chapter 29; Bucher & Grapes, 2011; Yardley, 1989). Primary sedimentary *calcite* (CaCO_3), *dolomite* ($\text{CaMg}(\text{CO}_3)_2$), and detrital *quartz* (SiO_2) react during progressive metamorphism to yield *diopside* ($\text{CaMgSi}_2\text{O}_6$), *forsterite* (Mg_2SiO_4), and *wollastonite* (CaSiO_3). Minor clay/shale impurities (illite) provide potassium and aluminum to generate *K-feldspar* via phlogopite breakdown. No other geological protolith possesses this combination of abundant Ca-Mg carbonates and calc-silicate phases.

(a) Mafic

INCORRECT

Why it is incorrect: A mafic protolith (metabasite such as basalt or gabbro) under granulite facies conditions produces a "two-pyroxene granulite" dominated by *orthopyroxene + clinopyroxene + calcic plagioclase ± garnet ± quartz* (as examined in Question 5). Mafic rocks have low CO₂ contents; primary calcite and dolomite do not exist in equilibrium in peak mafic granulites, nor does wollastonite form because calcium is strongly partitioned into plagioclase and clinopyroxene.

(b) Pelitic

INCORRECT

Why it is incorrect: Pelitic protoliths (shales, mudstones) are characterized by very high Al₂O₃ and FeO/MgO, with very low CaO. Under granulite facies, metapelites form assemblages of *sillimanite + garnet (almandine) + cordierite + biotite + K-feldspar + plagioclase + quartz*. Pelites completely lack the calcium required to form calcite, dolomite, diopside, or wollastonite.

(d) Ultramafic

INCORRECT

Why it is incorrect: While ultramafic rocks (peridotites, serpentinites) contain abundant Mg-olivine (forsterite) and minor pyroxene, they are extremely depleted in CaO and SiO₂. They never contain free quartz in equilibrium with forsterite, nor do they develop wollastonite (CaSiO_3) or significant calcite/dolomite in regional granulite facies terranes.



Synthesis Reference Tables, Diagnostic Criteria & High-Yield Practice

Protolith Chemical Fingerprints & 8 Exam-Grade Multiple-Choice Questions

Diagnostic Protolith Fingerprints Across Metamorphic Facies

Protolith Class	Sedimentary / Igneous Precursor	Key Chemical Characteristics	Granulite Facies Assemblage	Diagnostic Indicator Minerals
Carbonate / Calc-Silicate	Siliceous Dolomite, Limestone, Marl	High CaO, MgO, CO ₂ ; variable SiO ₂	Calcite + Dolomite + Diopside + Wollastonite + Forsterite ± Kfs	Wollastonite, Diopside, Forsterite in marble
Pelitic (Aluminous)	Shale, Claystone, Mudstone	High Al ₂ O ₃ , K ₂ O, FeO; low CaO	Sillimanite + Garnet + Cordierite + Kfs + Qtz ± Plag	Sillimanite, Kyanite, Staurolite, Cordierite
Mafic (Basic)	Basalt, Gabbro, Diabase	Moderate SiO ₂ , CaO, MgO, FeO; low K ₂ O	Orthopyroxene + Clinopyroxene + Plagioclase ± Garnet	Two Pyroxenes (Opx + Cpx)
Ultramafic	Peridotite, Dunite, Pyroxenite	Very high MgO, FeO; very low SiO ₂ , CaO, Al ₂ O ₃	Forsterite + Enstatite + Diopside + Spinel (no Quartz)	Abundant Forsterite, Anthophyllite, Serpentine
Quartzo-Feldspathic	Granite, Rhyolite, Arkose	High SiO ₂ , Na ₂ O, K ₂ O; low FeO, MgO, CaO	Quartz + K-feldspar + Plagioclase ± Garnet/ Biotite	Abundant Quartz & Feldspars, sparse mafics

8 High-Yield Practice Questions on Carbonate Metamorphism & Protolith Diagnosis

1. The reaction **Calcite + Quartz = Wollastonite + CO₂** is an example of which type of metamorphic reaction?

- (a) Continuous hydration (b) Discontinuous decarbonation (c) Polymorphic inversion (d) Oxidation reaction

Answer: (b) Discontinuous decarbonation — It is a univariant devolatilization reaction releasing CO₂ gas.

2. Why does wollastonite form at much lower temperatures in the presence of an H₂O-rich metamorphic fluid than in pure CO₂?

- (a) H₂O increases the activation energy
- (b) Dilution of the fluid phase by H₂O lowers the activity of CO₂ (a_{CO_2}), driving decarbonation to lower temperatures
- (c) H₂O transforms wollastonite into tremolite
- (d) H₂O reacts with quartz to form coesite

Answer: (b) — Reducing the mole fraction of CO₂ (X_{CO_2}) reduces the partial pressure and chemical potential of CO₂, shifting the equilibrium boundary toward lower temperatures.

3. In a pure dolomitic marble undergoing metamorphism, what mineral appears before diopside under moderate temperatures if quartz is present?

- (a) Tremolite
- (b) Sillimanite
- (c) Kyanite
- (d) Omphacite

Answer: (a) Tremolite — The reaction $5 \text{ Dolomite} + 8 \text{ Quartz} + 1 \text{ H}_2\text{O} = \text{Tremolite} + 3 \text{ Calcite} + 7 \text{ CO}_2$ introduces tremolite at lower amphibolite grade before diopside appears.

4. The presence of forsterite in a metamorphic marble indicates that:

- (a) The rock was originally an aluminum-rich bauxite
- (b) The protolith was dolomite-rich and quartz was completely consumed by earlier reactions
- (c) The metamorphism occurred at 0°C
- (d) The rock was subjected to hydrothermal gold mineralization

Answer: (b) — Forsterite forms by reaction between diopside and excess dolomite; it cannot coexist with quartz.

5. A metamorphosed rock consisting essentially of garnet (almandine-pyrope), sillimanite, cordierite, quartz, and biotite is derived from which protolith?

- (a) Basalt
- (b) Pelite (shale)
- (c) Limestone
- (d) Dunite

Answer: (b) Pelite (shale) — High Al and K with low Ca produce sillimanite, cordierite, and almandine garnet.

6. Which chemical component distinguishes a siliceous dolomite from a pure limestone?

- (a) Abundance of both MgO and SiO₂
- (b) Total absence of CaO
- (c) High Na₂O
- (d) High P₂O₅

Answer: (a) Abundance of both MgO and SiO₂ — Dolomite provides Mg and Ca, while detrital sand/chert provides silica.

7. The chemical formula of wollastonite is:

- (a) CaSiO₃ (b) CaMgSi₂O₆ (c) Mg₂SiO₄ (d) Ca₂Al₃Si₃O₁₂(OH)

Answer: (a) CaSiO₃ — Wollastonite is the calcium metasilicate pyroxenoid.

8. What is the role of potassium in the formation of K-feldspar in impure calc-silicate rocks?

- (a) It comes from detrital illite/muscovite and forms phlogopite, which breaks down to K-feldspar at high grades
- (b) It replaces silicon in the quartz lattice
- (c) It forms liquid native potassium metal
- (d) It acts as a catalyst without entering any mineral

Answer: (a) — Illite clay supplies potassium to form phlogopite at medium grades; at higher grades, phlogopite + calcite + quartz react to form K-feldspar + diopside.